AN IMMERSION OF ALGEBRAIC SPACES IS A MONOMORPHISM AND IS SEPARATED

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-morphisms.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof works through base changes to representable objects, which is exactly the definition of representability for morphisms of algebraic spaces.

Editorial note. The single reference of the counted spans is recorded as a non-load-bearing navigation pointer; the argument that consumes it stays inside the counted proof span.

Editorial note. Preamble: the three native excerpt spans of sources/stacks-spaces-morphisms/preamble.tex (lines 8-13, 14-18 and 154-217). The chapter file itself carries no preamble text.

Allowed standard prerequisites (editorial).

- Immersions of schemes are monomorphisms. Schemes, Lemma schemes-lemma-immersions-monomorphisms, used at proof lines 1986-1989. Gives the scheme case for every representable base change.
- A morphism is representable and a monomorphism if all base changes to schemes are. used at proof lines 1989-1990. Transfers the property to f.
- A monomorphism is separated. Lemma lemma-monomorphism-separated (used also at 2006-2009 in this chapter). Gives the 'in particular' clause.

Source reference. Original lemma 01L7: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2076–2076. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *An immersion of algebraic spaces is a monomorphism. In particular, any immersion is separated.*

Proof. Let $f : X \to Y$ be an immersion of algebraic spaces. For any morphism $Z \to Y$ with Z representable the base change $Z \times_Y X \to Z$ is an immersion of schemes, hence a monomorphism, see Schemes, Lemma 01L7. Hence f is representable, and a monomorphism. □

Date: stacks-project:spaces-morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.